

Baptiste PRAS

Master's student in Artificial Intelligence

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EDUCATION

Master's degree in Artificial Intelligence (PhD Track) <i>Université Paris-Saclay</i>	Sep 2025 – Present Orsay, France
Magistère in Computer Science (Honors Research Program) <i>Université Paris-Saclay</i> Graduated with Honors	Sep 2024 – Aug 2025 Orsay, France
Double Bachelor's Degree in Computer Science and Mathematics <i>Université Paris-Saclay</i>	Sep 2022 – Aug 2024 Orsay, France
Intensive English Language Program <i>EF New York</i> Achieved C2 Proficiency	Sep 2019 – May 2020 New York, USA

PUBLICATIONS

- Point-Based Counting of Cereals and Legumes in Intercropped Fields**
Junior Conference on Data Science & Engineering (Poster), 2026
- Fine-Grained Mention-Level Analysis of Biomedical Entity Linking Models**
Medical Informatics Europe (Full Paper), 2026
- Revisiting Optimal Class Ratios in Imbalanced Learning**
Junior Conference on Data Science & Engineering (Full Paper), 2025

PROFESSIONAL EXPERIENCE

- Research Internship** May 2026 – Aug 2026
Inria TAU Team - LISN Orsay, France
- Designed an automated per-species counting pipeline for wheat and pea in intercropped plots from UAV and smartphone imagery. Benchmarked few-shot and crowd-counting baselines with custom preprocessing: soil filtering, isotropic tiling, and leaf-tip point annotations. Worked with a fine-tuned point-query transformer and evaluated optimal scaling and edge filtering, reaching 4.69% MAPE with 74.2% F1 on wheat and 7.83% MAPE with 86.2% F1 on pea; presented the work as a poster at JDSE 2026.
- Contact: François Landes (LISN, Université Paris-Saclay)
- Supervised Research Project** Jan 2026 – Mar 2026
LISN - Laboratoire Interdisciplinaire des Sciences du Numérique Orsay, France
- Investigated Biomedical Entity Linking (BEL) strategies to normalize PICO entities (Population, Intervention, Comparison, Outcome) to the MeSH knowledge base. Developed and evaluated a hybrid normalization pipeline, comparing a custom rule-based script with a state-of-the-art graph-based entity linking model.
- Research Internship** May 2025 – Aug 2025
LISN - Laboratoire Interdisciplinaire des Sciences du Numérique Orsay, France
- Analyzed biomedical entity linking models on the BELB benchmark, focusing on generalization to rare or complex mentions. Developed quantitative and visual analyses of dataset characteristics (mention length, ambiguity, frequency) and their impact on prediction quality. Compared recent models, identified consistent weaknesses, and proposed improvements; published a paper in MIE 2026.
- Contact: Nona Naderi (LISN, Université Paris-Saclay)
- Supervised Research Project** Jan 2025 – May 2025
LISN - Laboratoire Interdisciplinaire des Sciences du Numérique Orsay, France
- Studied the impact of class imbalance on classification tasks using a spherical Teacher-Student perceptron. Conducted experiments in Python (Scikit-Learn, NumPy, Matplotlib) with different noise levels, loss functions, and training methods (GD, Langevin dynamics). Demonstrated that the optimal imbalance ratio in training sets differs from 0.5 in class-imbalanced problems; published a paper in JDSE 2025.

- Evaluated and refined LLM reasoning trajectories across complex coding and mathematical tasks under rigorous alignment protocols; designed adversarial prompts and assessed multi-step outputs to mitigate hallucinations and improve factual grounding for RLHF pipelines.

PROJECTS

Predictive Coding under Occlusion (Peekaboo, Ongoing) 🔗

- Built a controlled benchmark of digits moving behind occluders, with full ground truth, to test whether predictive coding networks keep track of hidden objects. Reimplemented PredNet in PyTorch and found why it never drew the digit (about 0.4% of the loss); weighting visible digit pixels by 10 cut next-frame MSE by 95% against a blank-frame baseline and 96% against copying the last frame, consistently across speeds and occlusion conditions.

Structure Detection in Fusion Plasma Simulations 🔗

- Ranked 4th out of 94 competitors (81% AP50) in detecting blob structures in fusion plasma simulation frames from a very small labeled set. Designed a two-round YOLOv8 pseudo-labeling pipeline, where an MLP on hand-crafted features (intensity statistics, Sobel gradients) selects the pseudo-labels and a patch CNN trained on the detector's own errors removes false positives.

Measuring Market Impact of Financial News 🔗

- In a team of four, built a frugal NLP pipeline converting financial news into structured (date, ticker, impact) events. Developed the hierarchical map-reduce summarizer (fine-tuned Flan-T5-large) and its LLM-as-a-judge audit of numeric fidelity and issuer grounding.

Scientific Citation Information Retrieval & Reranking 🔗

- In a team of three, built a multi-stage citation retrieval pipeline on 20,000 papers, improving MAP from 0.45 to 0.67 (+49%). Fused sparse lexical scores (BM25, citation contexts) with four dense encoders (UAE, BGE-large, E5, SciNCL) and metadata in an XGBoost ranker over 16 features.

Unpaired Image-to-Image Translation (CycleGAN from Scratch) 🔗

- Re-implemented CycleGAN in pure NumPy without autograd or PyTorch. Programmed manual analytical backward passes (im2col convolutions, instance norm, Adam) for ResNet generators and PatchGAN discriminators; validated cycle consistency ($L_1 \approx 0.20$) on horse2zebra and apple2orange.

SKILLS

Programming:

- Python (PyTorch, Scikit-Learn, NumPy), Bash, C/C++, Java, OCaml, SQL

Machine Learning & Vision:

- Deep Learning, Natural Language Processing, Computer Vision, Signal Processing

Tools:

- Linux, Slurm (Cluster Computing), Git, Hugging Face, OpenCV, L^AT_EX

ACHIEVEMENTS

Winner of an AI competition by designing the best-performing Dual Sudoku agent

3x Prologin finalist & 1x Gradient finalist, national algorithmic programming contests

LANGUAGES

French

Native

Russian

Conversational

English

Bilingual Proficiency

TOEFL iBT: 108/120

TOEIC: 990/990